

# Using OLS

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Session 3

# Goals

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- 1 How do we interpret and present OLS results?

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- 2 What inferences do we make in OLS?

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- 2 What inferences do we make in OLS?
- 3 How do we perform hypothesis tests in OLS?

# Properties of estimated parameters

- Intercept,  $\alpha$  or  $\beta_0$ : the conditional expectation of  $Y_i$  when  $X_i = 0$ 
  - ▶ Note, this isn't always of interest, for example when  $X = 0$  is impossible or far outside of the range of data. ( $X$  is blood pressure, or height etc.)
  - ▶ Also note that adding some quantity to  $X$ , let's say  $q$ , changes the intercept, but not the slope.
  - ▶ Often  $q$  is set to  $(-)\bar{X}$  so that the intercept is interpreted as the expected response at the average  $X$  value.

# Properties of estimated parameters

- Slope,  $\beta_1$ : average change in the mean of  $Y_i$  given a one-unit change in  $X_i$ .
  - ▶ Multiplication of  $X$  by a factor, let's say  $m$ , results in dividing the coefficient by a factor of  $m$ .
  - ▶ Dividing  $X$  by its sd results in the coefficient,  $\beta_1$  now represents a correlation coefficient.

# Cautionary Advice

- The magnitude of slopes (and intercepts) are meaningless without knowing the units to which they refer!
- The interpretation of the constant may be reasonable or it may not be. It may lie outside the bounds of what is observable.

# Table output

- Coefficient direction and size
- T-values and P-values
- N
- F-Test
- R-squared (adjusted  $R^2$ , BIC)
- Root MSE (the sd of the regression. The closer to zero, better the fit)
- Use commands such as *stargazer* or *modelsummary* to make tables look nice.

# Linear Regression Results

Call:

```
lm(formula = loggdp ~ risk, data = ajr2)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -1.8649 | -0.4604 | 0.1741 | 0.4682 | 1.1608 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 4.68741  | 0.41744    | 11.229  | < 2e-16 ***  |
| risk        | 0.51619  | 0.06252    | 8.257   | 1.42e-11 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7289 on 62 degrees of freedom

Multiple R-squared: 0.5237, Adjusted R-squared: 0.516

F-statistic: 68.17 on 1 and 62 DF, p-value: 1.421e-11

# Table output

## Model Summary Library

```
install.packages("modelsummary")  
library(modelsummary)  
  
# Fit your linear regression model  
model <- lm(formula = loggdp ~ risk, data = ajr2)  
  
# Create a summary table  
summary_table <- summary(model)  
  
# Print the summary table using modelsummary  
modelsummary(summary_table, stars = TRUE, output = "latex")
```

# Graph output

- Graphing results can provide more of an idea of substantive importance of results
- Easier to present (conferences, job talks)
- Should include measures of uncertainty (i.e. confidence intervals).
- ggplot2 or lattice are popular libraries in R
- modelsummary also produces plots

```
# Create a summary table using modelsummary
```

```
modelplot(model)
```

# Interpretation

Focus on three aspects in interpretation:

- Sign
  - ▶ Easy to see from table
- Substance
  - ▶ Hard to see from table
- Statistical significance
  - ▶ Easy to see from table, hard to understand

# Statistical Significance

Statistical Significance involves Null Hypothesis Testing

- Hypothesis testing is concerned with making decisions using data
- A null hypothesis is specified that represents the status quo, usually labeled  $H_0$ .
- Often, the null hypothesis is

$$E(Y|X) = E(Y)$$

- The null hypothesis is assumed true and statistical evidence is required to reject it in favor of a research or alternative hypothesis

# Hypothesis testing

- The alternative hypotheses are typically of the form  $<$ ,  $>$  or  $\neq$
- Note that there are four possible outcomes of our statistical decision process

| Truth | Decision              |                       |
|-------|-----------------------|-----------------------|
|       | Accept $H_0$          | Reject $H_0$          |
| $H_0$ | Correctly accept null | Type I error          |
| $H_a$ | Type II error         | Correctly reject null |

# Discussion

- Consider a court of law; the null hypothesis is that the defendant is innocent
- We require evidence to reject the null hypothesis (convict)
- If we require little evidence, then we would increase the percentage of innocent people convicted (type I errors); however we would also increase the percentage of guilty people convicted (correctly rejecting the null)
- If we require a lot of evidence, then we increase the percentage of innocent people let free (correctly accepting the null) while we would also increase the percentage of guilty people let free (type II errors)

# Hypothesis Testing in OLS

- We focus on a test statistic, usually  $\beta$ .
- We theorize about  $\beta$ 's sampling distribution.
- A sampling distribution tells us
  - 1 what values the statistic are likely to be
  - 2 how likely it is to assume those values prior to observing our data

# Roadmap: From Estimates to Inference

We know how to *estimate*  $\beta$ . But how *uncertain* is that estimate?

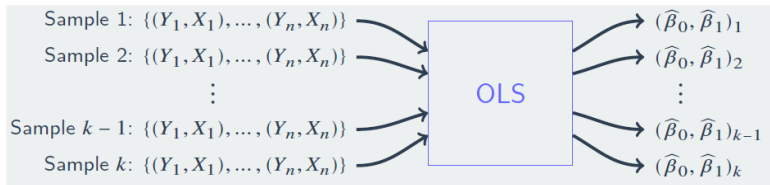
- To answer this, we need to understand three things:
  - 1 **Sampling distributions** — what would happen if we could repeat our study many times?
  - 2 **The standard error of  $\hat{\beta}$**  — how much does our estimate vary from sample to sample?
  - 3 **When we can treat the sampling distribution as normal** — the basis for our hypothesis tests and confidence intervals.

# What is a Sampling Distribution?

Imagine you could repeat your study thousands of times, each time drawing a new random sample from the same population and calculating  $\hat{\beta}$ .

- Each sample gives a *different* estimate of  $\beta$ .
- The **sampling distribution** is the distribution of all those possible  $\hat{\beta}$  values.
- It tells us how much our estimate would “bounce around” due to sampling variability alone.
- We never actually observe the sampling distribution — we only have one sample. But we can reason about it mathematically (and simulate it).

# Sampling distribution of the OLS estimator



- OLS is an estimator — a machine that we plug data into and get out estimates.
- Just like the sample mean or sample variance,  $\hat{\beta}$  has a sampling distribution with a center, a spread, and a shape.
- Our job now: figure out each of those properties.

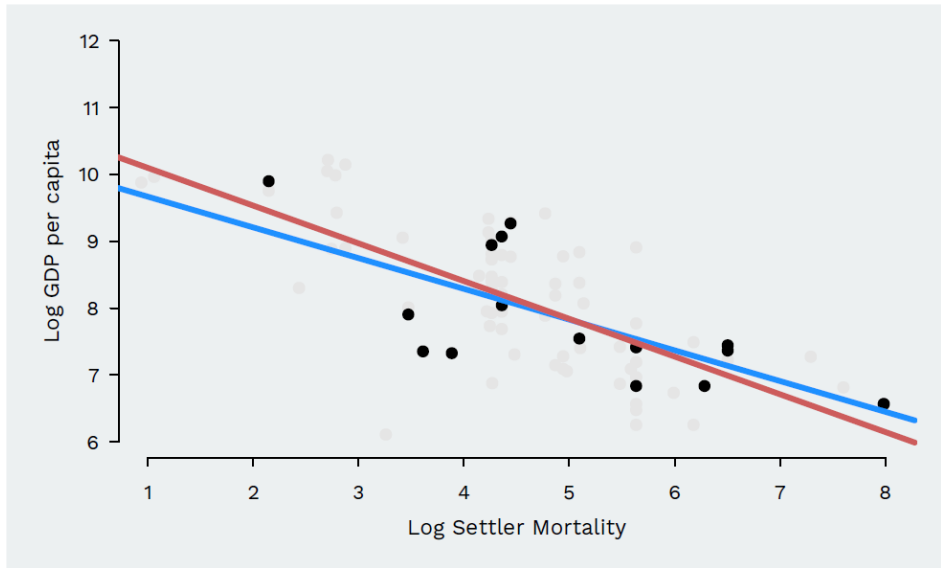
# Simulation procedure

- Let's use simulation to see the sampling distribution in action.
- Pretend that data from Acemoglu, Johnson, and Robinson's 2001 AER paper represents the population of interest.
- We'll draw a random sample of size  $n = 30$  with replacement.
- Use OLS to calculate the estimated slope and intercept.
- Repeat many times and see how the estimates vary.

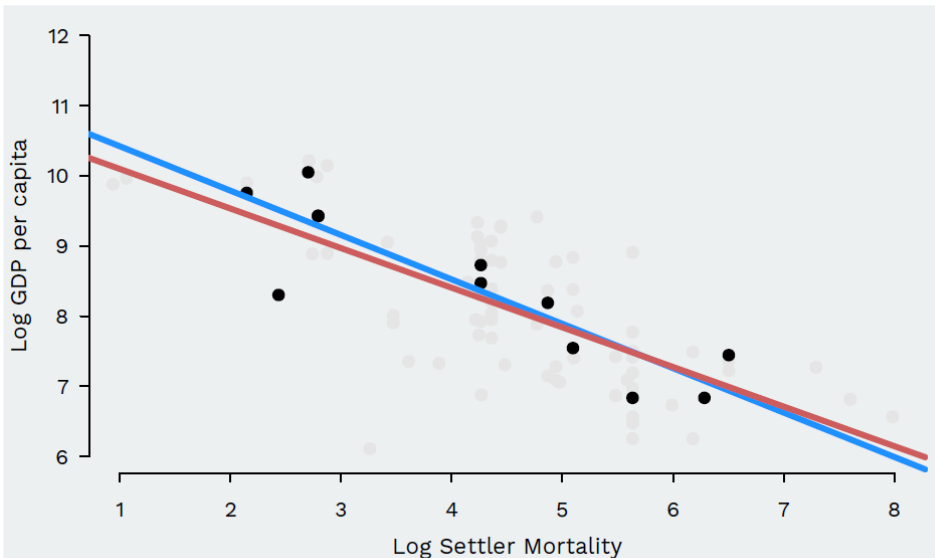
# Population Regression



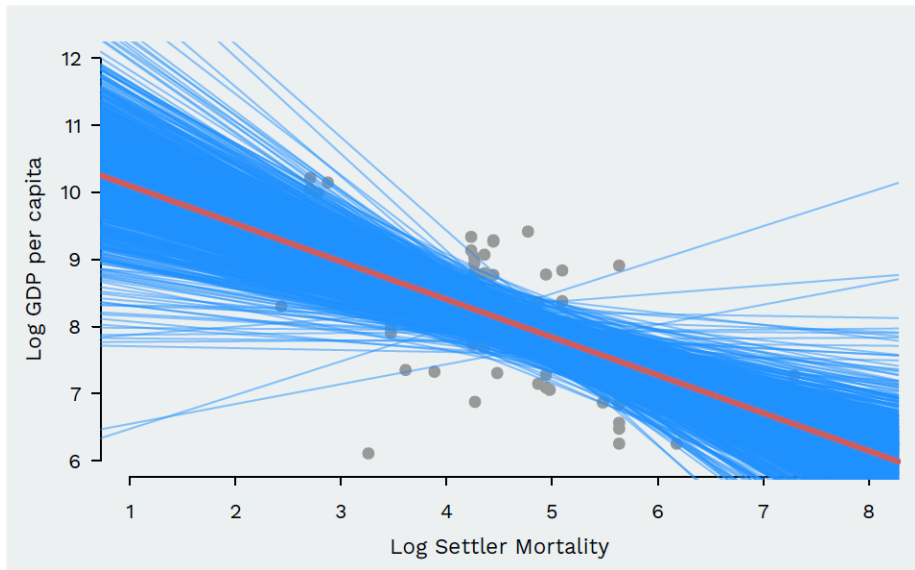
# Randomly sample from AJR



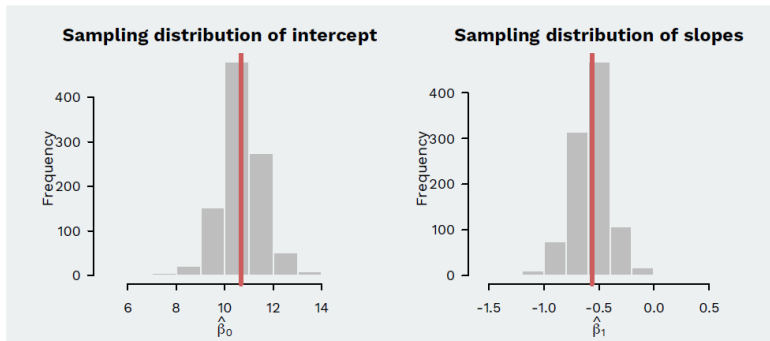
## Another random sample from AJR



# Many random samples from AJR



# Sampling distribution of the OLS estimator



- The estimated slopes vary from sample to sample, but they cluster around the true population slope.
- This clustering *is* the sampling distribution.

# Properties of the Sampling Distribution

Three questions about any sampling distribution:

- **Center** (Mean): Where is it centered?
  - ▶ Under OLS assumptions,  $E[\hat{\beta}] = \beta$ . The estimator is *unbiased*: on average, across repeated samples, we get the right answer.
- **Spread** (Variance / Standard Error): How much does  $\hat{\beta}$  bounce around?
  - ▶ This is what we need to derive — it depends on the variance of the errors and the spread of  $X$ .
- **Shape** (Distribution): What does the distribution look like?
  - ▶ Under certain conditions, approximately normal. The Central Limit Theorem (CLT) tells us why.

# The Central Limit Theorem (CLT)

The CLT is one of the most important results in statistics. It says:

- If you draw repeated random samples of size  $n$  from *any* population with mean  $\mu$  and finite variance  $\sigma^2$ , the distribution of sample means approaches a normal distribution as  $n$  grows large:

$$\bar{X} \xrightarrow{d} N\left(\mu, \frac{\sigma^2}{n}\right)$$

- This holds **regardless of the shape of the population distribution** — the original data can be skewed, bimodal, or otherwise non-normal.
- The larger the sample, the better the approximation.

# Why does the CLT matter for regression?

- Our OLS estimates  $\hat{\beta}$  are functions of the sample data — they are themselves sample statistics, like a sample mean.
- The CLT tells us that, in large enough samples, the sampling distribution of  $\hat{\beta}$  is approximately normal, centered on the true  $\beta$ .
- This is what we saw in the AJR simulation: the histogram of estimated slopes across many samples looked like a bell curve.
- Because we can treat the sampling distribution as approximately normal, we can build hypothesis tests and confidence intervals using the normal (or  $t$ ) distribution.
- **The CLT is basis for these inferential tools.**

## CLT: Key Takeaways

- The CLT applies to the sampling distribution of statistics (like  $\bar{X}$  or  $\hat{\beta}$ ) — *not* to the raw data themselves.
- It does *not* say your data are normally distributed. It says that if you could compute  $\hat{\beta}$  from many samples, those estimates would be approximately normally distributed.
- With small samples, the normal approximation may be poor — this is why we will use the  $t$ -distribution (which has heavier tails) rather than the standard normal for hypothesis testing.
- As  $n$  grows, the  $t$ -distribution converges to the standard normal.

# Sampling variance of estimated slope

We know that the sampling distribution is centered on the true population slope, but we don't know the population sampling variance.

$$V[\hat{\beta}_1] = ?$$

- Using an important assumption helps us derive an estimate for the variance of our slope.
- Homoskedasticity
- This refers to our assumptions about the error term

# What is the error term?

The error term represents the combined effect of the omitted variables, assuming that:

- the effect of the omitted variables is independent of each variable included in the equation
- the effect of the omitted variables is independent across subjects,
- the effect of the omitted variables has expectation 0
- Note: that the error term and residuals are often conflated. The error term is unknown, while the residuals are a function of our data and model choices.

# Homoskedasticity

The conditional variance of  $Y_i$  given  $X_i$  is constant:

$$V(Y_i|X_i = x) = V(\epsilon_i|X_i = x) = \sigma^2$$

- $V(Y_i|X_i = x)$  sometimes called the skedastic function, thus the name homoskedasticity.
- Under homoskedasticity

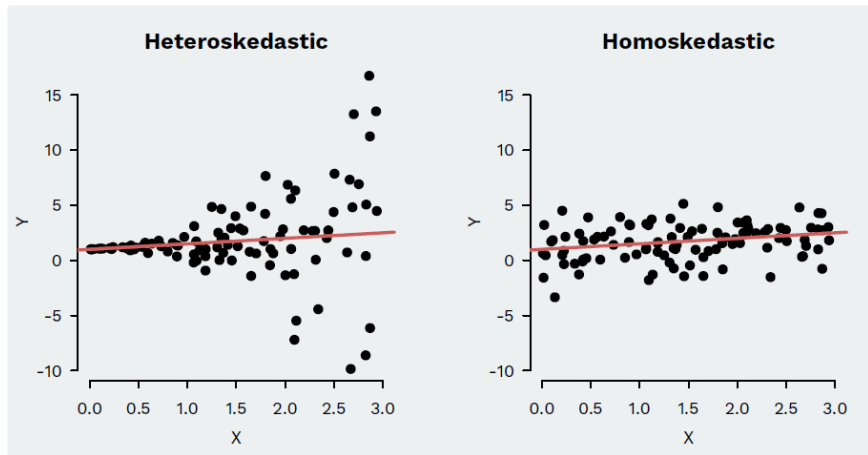
$$V[\hat{\beta}_1|X] = \frac{\sigma_\epsilon^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

- Standard error:

$$se[\hat{\beta}_1|X] = \sqrt{V[\hat{\beta}_1|X]} = \frac{\sigma_\epsilon}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

# Violations of homoskedasticity

Violations: magnitude of error differs at different levels of  $X$ . So our se's are wrong.



# Derive the sampling variance

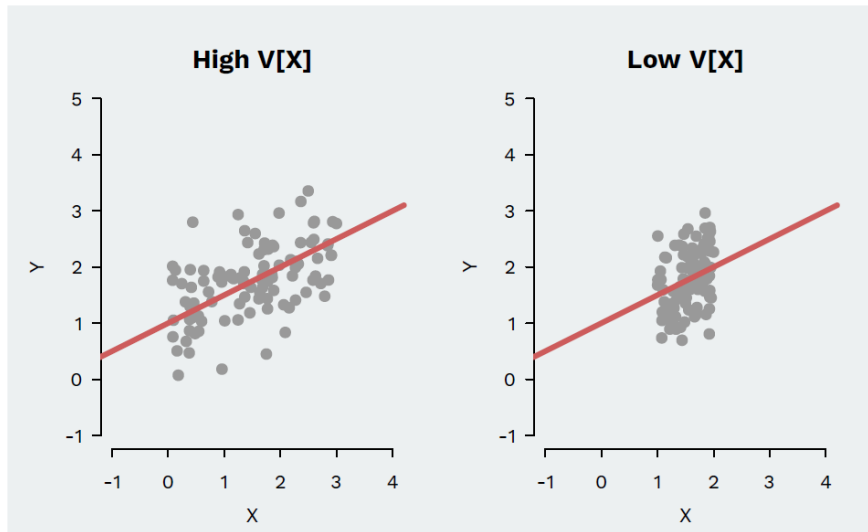
What drives the sampling variability of the OLS estimator?

# Derive the sampling variance

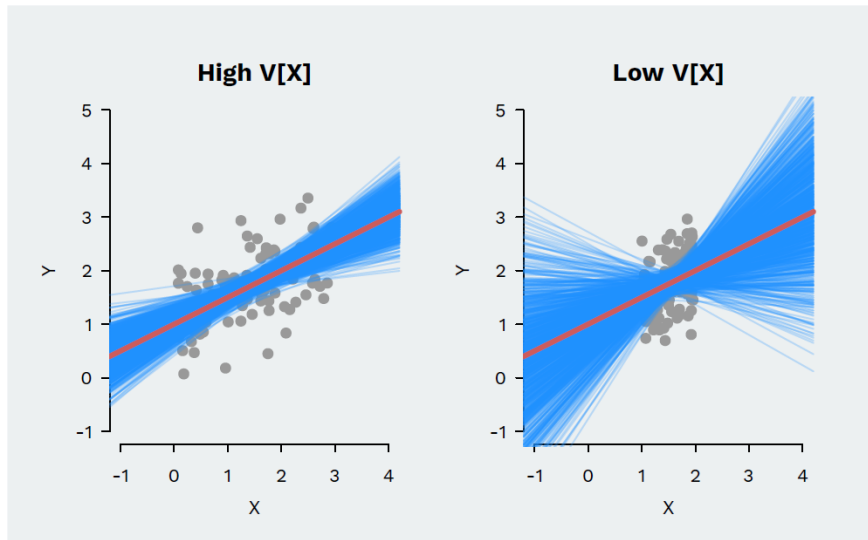
What drives the sampling variability of the OLS estimator?

- The higher the variance of  $Y$ , the higher the sampling variance
- The lower the variance of  $X$ , the higher the sampling variance
- As we increase  $n$ , the denominator gets large, while the numerator is fixed and so the sampling variance shrinks to 0.

# Variance in X and SEs



# Variance in $X$ and SEs



## Regression Uncertainty

- But we don't observe  $\sigma_\epsilon^2$  - it is the variance of the error (an unknown).
- We can estimate with residuals

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_i \hat{\epsilon}_i^2$$

- Why n-2? Because we are estimating two parameters (and thus lose 2 degrees of freedom). If we add another variable, and thus need to estimate another parameter, it would be n-3.
- Remember how that fits with the standard error of our slope

$$\widehat{\text{se}}(\hat{b}) = \sqrt{\frac{\hat{\sigma}^2}{\sum_i (x_i - \bar{x})^2}}$$

- If we wish to know whether our  $\beta$  is statistically significantly different from 0, our test statistic is  $\frac{\hat{\beta} - \beta}{\widehat{\text{se}}} \sim N(0, 1)$ .

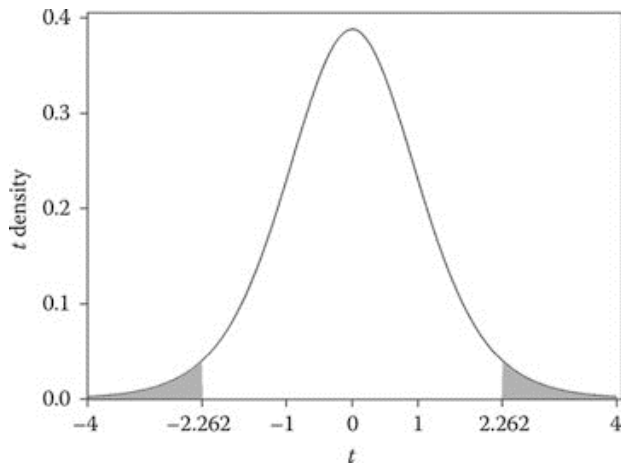
# Assumptions of Normality

- What can go wrong here is that the estimate of  $\sigma$  can be very inaccurate with small  $n$ .
- In practice, you have no way of knowing whether your estimated  $\sigma$  is too small.
- You can correct for the problem by properly accounting for the variability in your estimate of  $\sigma$  ; this is the purpose of using Student's t-distribution rather than the standard normal distribution.

# Student's t distribution

- The distribution of  $T$  uses Student's t-distribution with  $n - k - 1$  degrees of freedom, where  $k$  is the number of regressors. This distribution derives from both the standard normal and the chi-squared distributions.
- For simple (bivariate) regression,  $k = 1$ , so we have  $n - 2$  degrees of freedom.

## T distribution, $n=10$ , $\alpha = 0.05$



# T-test

Note that T-tests are only appropriate when the underlying data are normally distributed, or even are dealing with large samples (and we know the characteristics of our sample means and variance).

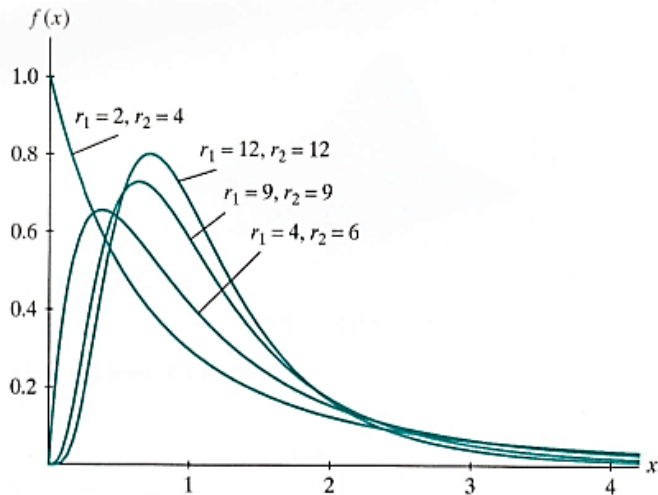
- Of course, we're probably violating these assumptions (All models are wrong), but we don't want to be too off base.
- If we violate these assumptions we can rely on other tests (more on this is a second).
- First, we need to examine these assumptions, either through theory, data exploration, or (hopefully) both.
- A QQ Plot is helpful in this case.

# F-Distribution

The null distribution of the ratio of variances estimated from independent normal samples. The F-distribution is defined similarly to chi-squared and Student's t-distributions

- If  $V_1 \sim \chi_{m1}^2$  and  $V_2 \sim \chi_{m2}^2$  independent of  $V_1$ , and if  $F = \frac{(V_1/m_1)}{(V_2/m_2)}$  then  $F \sim F_{m1,m2}$  is the F Distribution
- with  $m1$  numerator degrees of freedom and  $m2$  denominator degrees of freedom
- Note: Both the numerator and denominator have expectation 1.0, since the expected value of a chi-squared random variable is equal to its degrees of freedom.
- The center of the F-distribution is near 1.0, and it will converge to a center of 1 with more  $m1$  and  $m2$ .

# F Distribution



# F-Distribution

- Note the distributions all are centered around 1.0, though the mean is not necessarily 1.
- The distribution collapses around 1.0 in the case with many degrees of freedom - large values of  $m_1$  and  $m_2$  in the numerator and denominator
- Also notice that the plot for  $m_1 = 1$  is nearly identical to the chi-squared distribution with one degree of freedom

# F-Distribution

One application of the F-distribution that you can deduce is this: If a test statistic  $T$  has the  $T_m$  distribution, then  $T^2 \sim F_{1,m}$

$$T = \frac{Z}{\sqrt{V/m}}$$

where  $V \sim \chi_m^2$ . Now multiply each side by  $T$ .

$$T^2 = \frac{Z^2/1}{V/m}$$

where  $Z^2/1$  is in itself a variance.

$$T^2 \sim F_{1,m}$$

## F-Distribution and T distribution

Some software packages report F-statistics and some report t-statistics, even when the same models and hypotheses are considered. Here's why the p-values from the t-distribution and the F-distribution are identical.

$$p - value = Pr(T_m \geq |t| \text{ or } T_m \leq -|t|)$$

$$= Pr(T_m^2 \geq t^2)$$

by algebra

$$= Pr(F_{1,m} \geq t^2)$$

Thus, the two-tailed p-value from a t-test is equal to the upper-tailed p-value from the corresponding F-test, using  $t^2$  as the test statistic.

# F-Distribution

The most common use of the F-distribution is for testing a composite hypothesis that states that many hypotheses are simultaneously true.

- Suppose that you have an experiment where you want to study the effects of three different treatments simultaneously or examine one treatment that that categorically different values.
- At the end of the study, there will be differences in the patient outcomes between the three groups, but are these differences explainable by chance alone?
- The analysis-of-variance model, or **ANOVA** model, is used to analyze such multiple-group data.

# Confidence intervals

Confidence intervals provide a way to add more information to our estimators, leveraging what we now know from the CLT.

- We can find an interval  $(U,L)$  that we think has a high probability of containing our true estimand (i.e. the true value of the population value that we want to know).
- The size of the interval gives us an idea of how close we can estimate our estimand.

# Confidence intervals

- As we established, the CLT tells us the sample mean,  $\bar{X}$ , is approximately normal with mean  $\mu$  and standard error  $\sigma/\sqrt{n}$
- $\mu + 2\sigma/\sqrt{n}$  is pretty far out in the tail (only 2.5% of a normal being larger than 2 sds in the tail)
- Similarly,  $\mu - 2\sigma/\sqrt{n}$  is pretty far in the left tail (only 2.5% chance of a normal being smaller than 2 sds in the tail)
- Using this information, we can derive an interval estimate of  $\theta$ , consisting of two bounds (upper and lower) within which we expect  $\theta$  to reside

# Confidence intervals

- 95% confidence intervals refers to the fact that if one were to repeatedly get samples of size  $n$ , about 95% of the intervals obtained would contain  $\mu$
- Adding and subtracting 2 SEs from our test statistics derives for 95% intervals
- Confidence intervals get wider as the coverage increases (i.e. 99%)
- Confidence intervals get narrower with larger sample sizes (why?)

# Confidence intervals

Confidence intervals are tricky to interpret and controversial for methodologists.

- CI's are at the fore-front of the frequentists-Bayesian debates
- CI's are a frequentists construction
- A general accepted interpretation is this – If we repeated an experiment many times, and we constructed a confidence interval for each sample from the experiment, then some percent of the time (i.e. 95 percent), the CI would contain the estimand.

# Confidence intervals

- Note that the percent (i.e. 95 percent) is not referring the CI itself, but rather to the confidence in the method that is used to obtain the interval. In other words, the CI does not refer to a specific sample (i.e. the one we actually draw), but to the idea of repeated sampling (think CLT).
- This leads to divergent treatments of CI's.
- Correct interpretation: across 95% of random samples, the constructed confidence interval will contain the true value.

# Confidence intervals

- On one hand, some would argue that we have no idea if the CI calculated for a specific sample actually contains the true population parameter of interest.
- People who take this view, advocate the use of the credibility interval

$$P(\theta_L \leq \theta \leq \theta_U) = 0.95$$

- On the other hand, from CLT we know the distribution from which our estimates are drawn, which should indicate the distribution from which our CI's are drawn.

# Confidence intervals

However, all that is for methodologists to hash out. In practice, it is useful to become familiar with CI's and utilize them in your research.

- The use of confidence intervals is an established norm in graphing effects and marginal effects.
- We often see CI's in tables instead of standard errors.
- CI's can indicate the plausible ranges of effects sizes (i.e small or large) and can give indications of significance (does CI overlap with zero?).

# Regression Uncertainty

- We can create 95% confidence intervals around our estimates of  $\alpha$  and  $\beta$ .
- We can test whether  $\beta$  is statistically significant using these confidence intervals